is riddled with problems relating to the airframe, RD MK-33 engine and fly-by-wire system," the report had said. This ruled out the MiG-29k for service in the navy's second, newest aircraft carrier, the INS *Vikrant*.

In 2019, the Indian Navy initiated the process to acquire 57 multi-role combat aircraft for its carriers. Simultaneously, India's indigenous Twin Engine Deck Based Fighter (TEDBF) is being developed by the Aeronautical Development Agency (ADA), a department of the Defence Research and Development Organisation (DRDO). But it is still in the research and development stage, with the first test flight expected by 2026 and induction by 2031. Till that time, the navy decided to get 26 fighter aircraft for INS *Vikrant* as a 'stopgap arrangement'. Twelve Russian-origin MiG-29Ks are likely to be deployed on INS *Vikrant* till the Rafale-Ms join the naval fleet.

After sustained trials of the Rafale-M and the bigger Boeing F/A-18 Super Hornet, the navy in its report to the MoD in December 2022 stated that the French fighter jet was found to be "more suitable in meeting the operational requirements and criteria". Naval sources claim that the F-18 had difficulty fitting into the INS *Vikrant*'s elevator even with its foldable wings. This was a big operational handicap. Though the Rafale-M does not have foldable wings, Dassault has agreed to make design changes—like tweaking the pylon structure of the aircraft—to make it suitable for the *Vikrant*. This enabled it to clear the Navy's technical parameters.

AN ALL-ROUND MACHINE

Rafale-M is a twin-engine, single-seat aircraft capable of undertaking 'quick reaction alert', air defence and air policing missions, air support for ground forces and reconnaissance missions. It can also be used for deep strikes in enemy territory, for projecting power and for nuclear deterrence. The Rafale-M can carry out air-to-ground as well as air-to-air strikes during the same sortie. Its "service ceiling"—the altitude at which an aircraft can no longer rise faster than 100 feet per minute under standard air conditions—stands at 50,000 feet.

Experts say that naval jets possess specialised avionics, sensors, and communication equipment for maritime/ carrier operations. According to Dassault, Rafale-M has specific capabilities like the RBE2

RAFALE-M: NAVY'S FORCE MULTIPLIER



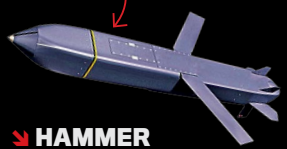
Selected for service on India's indigenous aircraft carrier INS *Vikrant*, the powerful and agile Rafale Marine will add capabilities in the Navy's strike, air defence and reconnaissance missions

WEAPONS SYSTEMS

METEOR Long-range air-to-air missile

MICA Air-to-air Beyond Visual Range (BVR) interception, combat and self-defence missiles

SCALP Long-range stand-off missile



HAMMER (Highly Agile and Manoeuvrable Munition Extended Range) modular, rocket-boosted air-to-ground precision-guided weapon series

AM39 EXOCET Anti-ship missile

Laser-guided bombs

AIRCRAFT ON INS VIKRAMADITYA, INDIA'S OPERATIONAL CARRIER

MIKOYAN MIG-29K 38
Multi-role fighter jet



KAMOV KA-31 10
Airborne early warning and control helicopters





INS *VIKRANT* Ready for Rafale

SINCE THE IAF'S RAFALES **SHARE 80 PER CENT OF COMPONENTS WITH RAFALE-Ms,** THE NAVY DEAL WOULD HELP SAVE COSTS ON MAINTENANCE